A2: Equity Plotting

An **equity curve** shows how a simulated portfolio's value changes over the dates in a backtest. Plotting several equity curves on the same axes makes it easier to compare strategies than reading their final values alone.

Plotting is an optional development aid, not a submission requirement. Use it after your backtester can write the equity CSV described in R6.

The plotting tool is included in the optional **a2-tools.zip (https://canvas.cornell.edu/courses/90271/files/16180858?wrap=1)** **(https://canvas.cornell.edu/courses/90271/files/16180858/download?download_frd=1)** bundle. Follow the bundle's README for installation and setup, including how to make `tools/plot_equity` available in your project. Run the commands below from the top-level directory of your A2 project.

Produce comparable equity CSVs

A useful comparison changes the strategy while holding the other inputs constant. In particular, run every strategy against the **same market data** and normally give every strategy the **same initial cash**. Otherwise, a difference between two curves might be caused by different inputs rather than different strategy decisions.

The following example runs the three required strategies against `medium-tech-2007.csv`, starting each with `$10,000.00`. First create a directory for the generated files:

```
mkdir -p plots
```

Run the cash strategy:

```
dune exec bin/main.exe -- data/medium-tech-2007.csv \
  cash plots/cash.csv --cash 10000.00
```

Run buy-and-hold for AAPL:

```
dune exec bin/main.exe -- data/medium-tech-2007.csv \
  buy-and-hold plots/buy-and-hold-AAPL.csv \
  --cash 10000.00 --hold AAPL
```

Run the default stateful momentum strategy:

```
dune exec bin/main.exe -- data/medium-tech-2007.csv \
  momentum plots/momentum.csv --cash 10000.00
```

If you implemented a different stateful strategy, substitute its strategy name and any options it requires. Give each equity CSV a descriptive filename. The plotting tool uses the filename—not the `value` column header—as that run's legend label. Including `AAPL` in `buy-and-hold-AAPL.csv`, for example, records which security that run held.

Plot the comparison

Pass `tools/plot_equity` the data-set description, all the equity CSVs, and finally the chart filename:

```
tools/plot_equity --data-set "tech-stocks-2007" \
  plots/cash.csv \
  plots/buy-and-hold-AAPL.csv \
  plots/momentum.csv \
  plots/strategy-comparison.svg
```

The `--data-set` value is a short description printed in the chart title. It does not name a file for the tool to open. The plotting tool cannot verify which input file produced an equity CSV, so you are responsible for supplying curves derived from the same market data. The x axis already shows the date range; the description should identify the data set without repeating those dates.

The final positional argument is always the chart to create. Its extension selects the output format:

- `.svg` produces a scalable vector image and is the recommended default.
- `.png` produces a bitmap image that is convenient to share.
- `.pdf` produces a document suitable for reports.

For example, change the last argument to `plots/strategy-comparison.png` to produce a PNG instead.

Read the chart

The horizontal axis shows input dates, and the vertical axis shows total portfolio value in dollars after that date's orders have been processed. The legend associates each curve with the basename of its equity CSV. The title identifies the common market data, and the subtitle identifies the course.

A higher final point means a higher final portfolio value for that particular historical run. Also inspect what happened between the first and last dates: two strategies can finish near the same value while following very different paths. A historical comparison is useful for understanding program behavior; it is not evidence that a strategy will perform similarly in the future.

Plot one file or a prepared comparison file

The tool can also plot a single R6 equity CSV:

```
tools/plot_equity --data-set "tech-stocks-2007" \
  plots/momentum.csv plots/momentum.svg
```

When given one CSV, the tool plots every value column after `date` and uses those column headers as legend labels. This also permits a separately prepared comparison CSV with several value columns. When given multiple CSVs, however, each one must have exactly the R6 header `date,value`.

Troubleshooting

- Run `tools/plot_equity --help` to see the accepted command-line shape.
- If the shell reports that Gnuplot is missing, follow the installation instructions in the tools bundle's README.
- An error saying that a comparison input must begin with `date,value` usually means you supplied a CSV of market data instead of an equity CSV, or your R6 output has the wrong header.
- An unreadable-input error usually means the filename is misspelled or the backtest did not create the expected CSV.
- If a legend label is unclear, rename the corresponding equity CSV and run the plotting command again.